

Q1. Explain why publishing and sharing reports is considered more important than just building dashboards in Power BI Desktop?

Ans: Publishing and sharing reports in Power BI is more important than just building dashboards in Power BI Desktop because the real value of business intelligence comes from **using and distributing insights**, not just creating them.

Here's the reasoning clearly:

1. Collaboration and Accessibility

Power BI Desktop is mainly for development. When you publish reports to the Power BI Service, others (team members, managers, stakeholders) can access them from anywhere via web or mobile. This ensures insights are **shared across the organization**, not locked on one person's computer.

2. Decision-Making

Dashboards are only useful if decision-makers can see them. Publishing allows stakeholders to interact with reports, filter data, and gain insights in real time, which supports **faster and better decision-making**.

3. Real-Time Data & Updates

Published reports can be connected to live data sources and scheduled refreshes. This ensures users always see **up-to-date information**, unlike static reports in Desktop.

4. Security and Control

Power BI Service provides features like **role-based access control, row-level security (RLS), and permissions**, ensuring that the right people see the right data.

5. Sharing and Distribution Features

Once published, reports can be:

- Shared via links
- Embedded in apps or websites
- Distributed through Power BI Apps
- Subscribed to via email updates

This makes insights **easy to distribute at scale**.

6. Collaboration & Version Control

Teams can work together on reports in shared workspaces, reducing duplication and ensuring everyone works on the **latest version**.

7. Integration with Other Tools

Published reports integrate with tools like **Microsoft Teams, SharePoint, and Excel**, making them part of daily workflows.

Q2. Define Power BI Service and explain its role in comparison to Power BI Desktop in the reporting lifecycle?

Ans: Power BI Service (also called *PowerBI.com*) is an online environment where users can:

- **Host reports and dashboards**
 - **Share insights with others**
 - **Set up data refresh and alerts**
 - **Collaborate through workspaces**
-

Role in Comparison to Power BI Desktop

The reporting lifecycle in Power BI typically has two main stages:

1. Power BI Desktop (Development Stage)

Power BI Desktop is used for:

- Data connection and transformation
- Data modeling (relationships, DAX calculations)
- Report and dashboard creation

It is mainly a development tool used by analysts.

2. Power BI Service (Deployment & Consumption Stage)

Power BI Service is used after reports are created. Its role includes:

a. Publishing Reports

Reports built in Desktop are uploaded to the Service so others can access them.

b. Sharing and Collaboration

Users can share dashboards with teams, create workspaces, and collaborate in real time.

c. Data Refresh & Real-Time Access

Supports scheduled refresh and live connections so reports stay up to date.

d. Security Management

Implements permissions and Row-Level Security (RLS) to control data access.

e. Dashboard Creation

Allows combining visuals from multiple reports into interactive dashboards.

f. Integration & Accessibility

Accessible via browsers and mobile apps; integrates with tools like Teams and SharePoint.

Q3. What does publishing a report mean in Power BI? List any two outcomes that occur after a report is published?

Two outcomes after publishing a report

1. Report becomes accessible online

Once published, the report is available in the Power BI Service, where users can:

- View it through a web browser or mobile app
- Interact with visuals (filters, slicers, drill-downs)

This makes the report accessible beyond the local system.

2. Sharing and collaboration become possible

After publishing, you can:

- Share the report with team members
- Add it to workspaces or apps
- Control access using permissions

This enables collaboration and organization-wide usage.

Q4. List and explain the steps involved in publishing a report from Power BI Desktop to Power BI Service in the correct sequence?

Ans: Steps to Publish a Report

1. Prepare and Save the Report

- Create report in Power BI Desktop
- Ensure all visuals, data models, and calculations are correct
- Save the file (.pbix format)

2. Sign in to Power BI Account

- Open Power BI Desktop
- Sign in using your organizational (work/school) account

This connects Desktop to the cloud service.

3. Click on the “Publish” Button

- Go to the Home tab
- Click Publish

This initiates the upload process.

4. Select a Destination Workspace

- Choose where you want to publish the report (e.g., *My Workspace* or a shared workspace)

Workspaces help organize and manage reports.

5. Confirm Publishing

- Click Select/OK to start publishing
- Wait for the process to complete

A success message will appear once done.

6. View Report in Power BI Service

- Click the link provided after publishing, or
- Go to the Power BI Service in your browser
- The report (interactive visuals)
- The dataset (data model behind it)

7. Configure Post-Publishing Settings (Optional but Important)

After publishing:

- Set up scheduled data refresh
- Apply Row-Level Security (RLS)
- Share the report with others or create dashboards

Q5. Differentiate between Report and Dashboard in Power BI based on structure and usage?

Ans: Report: A report is a detailed, multi-page collection of visuals that allows users to explore and analyze data deeply. Users can interact with filters and drill into data to uncover insights.

Dashboard:

A dashboard is a single-page overview that displays key information using pinned visuals from one or more reports. It is designed for quick monitoring and decision-making.

In Short

- Report = Detailed analysis tool
- Dashboard = Quick summary view

Q6. Explain the difference between the following three sharing methods in Power BI Service:

Direct Report Sharing, Workspace Access, Power BI Apps Also mention one use case for each?

Ans: 1. Direct Report Sharing

Sharing a report directly with specific users by granting them access via email or link.

Key Features:

- Simple and quick sharing
- Access limited to selected individuals
- Users can view (and sometimes interact with) the report

Use Case:

Sharing a sales performance report with a specific manager or small team for quick review.

2. Workspace Access:

Providing users access to a workspace where multiple reports, dashboards, and datasets are stored.

Key Features:

- Supports collaboration among team members
- Different roles (Admin, Member, Contributor, Viewer)
- Users can create, edit, and manage content (based on role)

Use Case:

A data analytics team collaborating to build and maintain reports for an entire department.

3. Power BI Apps

Packaging and distributing a collection of reports and dashboards to a large audience in a structured way.

Key Features:

- Read-only access for end users
- Clean, user-friendly experience
- Ideal for large-scale distribution
- Centralized updates by report creators

Use Case:

Distributing a company-wide dashboard app (e.g., finance or HR reports) to all employees.

Q7. Describe the four workspace roles available in Power BI and explain how access control is managed using these roles?

Ans: Four Workspace Roles in Power BI

1. Admin

Description: Full control over the workspace

Permissions:

- Add/remove users
- Assign roles
- Publish, edit, and delete content
- Configure settings (e.g., security, app publishing)

Use Case: Team lead or BI manager managing the entire workspace.

2. Member

Description: Can collaborate and manage most content

Permissions:

- Publish and edit reports/dashboards
- Share content
- View and interact with all items
- Cannot manage workspace settings like Admins

Use Case: Core team members actively working on reports.

3. Contributor

Description: Can create and modify content but with limited control

Permissions:

- Add and edit reports and datasets
- Cannot share content externally
- Cannot manage user access

Use Case: Report developers who build content but don't manage access.

4. Viewer

Description: Read-only access

Permissions:

- View and interact with reports and dashboards
- Cannot edit or share content

Use Case: Business users or stakeholders who only consume reports.

How Access Control is Managed

Power BI uses these roles to implement Role-Based Access Control (RBAC):

- Permission Assignment: Users are assigned roles based on their responsibilities
- Controlled Actions: Each role limits what actions a user can perform (view, edit, share, manage)
- Security: Prevents unauthorized changes or access to sensitive data
- Collaboration Balance: Ensures the right people can contribute while others only consume

For example:

- Admin controls who gets access
- Members and Contributors build content
- Viewers consume insights safely

In Short

- Admin → Full control
- Member → Collaborate & manage content
- Contributor → Create content
- Viewer → Read-only access
-

Q8. Why are Power BI Apps considered more secure and scalable than direct sharing? Explain this from a governance and enterprise usage perspective?

1. Governance (Control & Management)

Centralized Control

- Power BI Apps are managed from a single workspace
- Only authorized users (Admins/Members) can publish or update the app

This ensures strict control over what content is released

Version Control

- Updates are made once and automatically reflected for all users
- Prevents multiple versions of the same report (a common issue in direct sharing)

Ensures a single source of truth

Permission Management

- Access is granted to the app, not individual reports
- Works with organizational security (e.g., groups, roles)

Reduces risk of accidental over-sharing

2. Security Advantages

Read-Only Access

- Users typically get view-only access in apps
- They cannot modify or reshare reports

Protects data integrity and prevents misuse

Controlled Distribution

- Unlike direct sharing (which can be forwarded), apps are distributed to defined audiences

Minimizes data leakage risks

Integration with Security Policies

- Supports features like Row-Level Security (RLS)
- Works with enterprise identity systems (e.g., Azure Active Directory)

Ensures data is visible only to authorized users

3. Scalability (Enterprise Usage)

Designed for Large Audiences

- A single app can be distributed to hundreds or thousands of users

Direct sharing becomes inefficient at this scale

Simplified User Experience

- Users access all relevant reports in one packaged app
- No need to manage multiple links or permissions

Improves usability and adoption

Efficient Maintenance

- Admins update content in one place instead of managing access report-by-report

Saves time and reduces administrative overhead

Q9. Explain Row-Level Security (RLS) in Power BI.

How does RLS ensure that different users see different data using the same report?

Ans: **Row-Level Security (RLS)** in Power BI is a feature that **restricts data access at the row level**, so different users can see **only the data they are authorized to view**, even when using the same report.

Example Scenario:

A company has a sales report with data from multiple regions.

- **Manager A (India)** → sees only India sales
- **Manager B (USA)** → sees only USA sales

Even though:

- They open the **same report**
- With the **same visuals**

The data shown is filtered based on their role.

Q10. Assume an organization shares reports without using Apps, workspace roles, or RLS.

Discuss any three risks or issues that may arise related to security, governance, or data misuse, and explain how Power BI best practices help prevent them?

Ans: 1. Unauthorized Data Access (Security Risk)

Problem:

- Without RLS or controlled access, all users may see full datasets, including sensitive information (e.g., salaries, confidential sales data).
- Reports shared directly can be forwarded, leading to data exposure beyond intended users.

Impact:

- Data breaches
- Compliance violations (e.g., privacy regulations)

Best Practice Solution:

- Implement Row-Level Security (RLS) to restrict data visibility
- Use Power BI Apps to distribute reports to specific audiences

- Assign workspace roles to control who can view or edit content

Ensures users only see authorized data

2. Lack of Governance and Version Control

Problem:

- Multiple copies of reports may circulate through direct sharing
- No central control over updates or changes
- Users may rely on outdated or inconsistent reports

Impact:

- Confusion and conflicting insights
- Poor decision-making due to incorrect data

Best Practice Solution:

- Use Power BI Apps for centralized distribution
- Maintain reports in controlled workspaces
- Ensure a single source of truth with managed updates

Guarantees consistency and reliability of reports

3. Data Misuse and Uncontrolled Modifications

Problem:

- Users with excessive access may edit, duplicate, or misuse reports
- Lack of role-based control increases chances of accidental or intentional changes

Impact:

- Loss of data integrity
- Incorrect reporting and analysis

Best Practice Solution:

- Assign appropriate workspace roles (Admin, Member, Contributor, Viewer)
- Restrict editing rights to authorized users only
- Provide read-only access via Apps for most users

Protects report structure and ensures controlled usage

Conclusion

Without proper controls, organizations face:

- Security breaches
- Poor governance
- Data misuse risks

By following Power BI best practices—using RLS, workspace roles, and Apps—organizations can ensure:

- Secure data access
- Centralized governance
- Reliable and controlled reporting